

Supplementary Material: Positive-Class Weighting Induces Systematic Probability Distortion in Deep Learning Models for Chest Radiography under Deidentified-Date Cohort Shift

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Section S1: Comprehensive Cross-Institutional Factorial Matrix

Table S1 provides the complete $2 \times 2 \times 4$ cross-institutional factorial matrix on the official MIMIC-CXR-JPG test cohort ($N = 3,403$ frontal radiographs from 3,041 diagnostic studies across $N = 289$ unique patients). Both unweighted and positive-weighted binary cross-entropy (BCE) models are evaluated across DenseNet-121 and ResNet-50 architectures for all four thoracic abnormalities. For positive-weighted models, paired zero-parameter analytic logit recalibration ($z \mapsto z - \log w$) is evaluated alongside patient-clustered 1,000-replicate bootstrap 95% confidence intervals.

Section S2: Diagnostic Study-Level Sensitivity Analysis

To evaluate sensitivity to repeated within-patient examinations and multi-image studies, Table S2 summarizes external validation performance aggregated strictly at the diagnostic study level ($N = 3,041$ studies across $N = 289$ unique patients). In accordance with standard radiological diagnostic practice, multiple views acquired during a single examination are aggregated using maximum probability pooling ($\max_i p_i$, where detection on any view indicates study-level positive status). Supplementary Table S2 evaluates

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Table S1: Supplementary Table S1: Complete 16-Row Cross-Institutional External Validation Matrix on the Official MIMIC-CXR-JPG Frontal Test Cohort ($N = 3,403$ Radiographs, 3,041 Studies, 289 Unique Patients). Evaluated on 3-Seed Probability Ensembles with Image Prevalence (π), Empirical Null Reference Baselines ($\text{Brier}_{\text{null}} = \pi(1 - \pi)$), Patient-Clustered Bootstrap 95% Confidence Intervals, Brier Skill Scores ($\text{BSS} = 1 - \text{Brier}/\text{Brier}_{\text{null}}$), and Paired Analytic Recalibration ($\Delta\text{Brier} = \text{Brier}_{\text{corr}} - \text{Brier}_{\text{raw}}$).

Architecture	Objective	Pathology	Prevalence (π)	$\text{Brier}_{\text{null}}$	AUROC [95% CI]	Brier (Raw $\rightarrow$ Corr)	ΔBrier [95% CI]	BSS (Raw $\rightarrow$ Corr)
DenseNet-121	Unweighted BCE	Pleural Effusion	32.18%	0.2182	0.688 [0.656, 0.717]	0.2488	–	-0.140
DenseNet-121	Unweighted BCE	Cardiomegaly	26.33%	0.1940	0.523 [0.491, 0.556]	0.2375	–	-0.224
DenseNet-121	Unweighted BCE	Pneumonia	10.05%	0.0904	0.607 [0.560, 0.650]	0.0986	–	-0.091
DenseNet-121	Unweighted BCE	Edema	21.33%	0.1678	0.562 [0.524, 0.596]	0.2124	–	-0.266
DenseNet-121	Pos-Weighted BCE	Pleural Effusion	32.18%	0.2182	0.668 [0.636, 0.700]	0.2536 $\rightarrow$ 0.2451	-0.0084 [-0.0401, +0.0240]	-0.162 $\rightarrow$ -0.123
DenseNet-121	Pos-Weighted BCE	Cardiomegaly	26.33%	0.1940	0.525 [0.492, 0.558]	0.2907 $\rightarrow$ 0.2328	-0.0579 [-0.0790, -0.0349]	-0.499 $\rightarrow$ -0.200
DenseNet-121	Pos-Weighted BCE	Pneumonia	10.05%	0.0904	0.598 [0.553, 0.639]	0.1078 $\rightarrow$ 0.0983	-0.0095 [-0.0170, -0.0016]	-0.192 $\rightarrow$ -0.087
DenseNet-121	Pos-Weighted BCE	Edema	21.33%	0.1678	0.581 [0.549, 0.611]	0.1914 $\rightarrow$ 0.2131	+0.0217 [+0.0118, +0.0307]	-0.141 $\rightarrow$ -0.270
ResNet-50	Unweighted BCE	Pleural Effusion	32.18%	0.2182	0.677 [0.645, 0.709]	0.2430	–	-0.113
ResNet-50	Unweighted BCE	Cardiomegaly	26.33%	0.1940	0.519 [0.489, 0.548]	0.2333	–	-0.203
ResNet-50	Unweighted BCE	Pneumonia	10.05%	0.0904	0.583 [0.540, 0.623]	0.0982	–	-0.086
ResNet-50	Unweighted BCE	Edema	21.33%	0.1678	0.516 [0.480, 0.552]	0.2126	–	-0.266
ResNet-50	Pos-Weighted BCE	Pleural Effusion	32.18%	0.2182	0.620 [0.588, 0.653]	0.2665 $\rightarrow$ 0.2557	-0.0109 [-0.0417, +0.0253]	-0.221 $\rightarrow$ -0.172
ResNet-50	Pos-Weighted BCE	Cardiomegaly	26.33%	0.1940	0.522 [0.491, 0.552]	0.2822 $\rightarrow$ 0.2211	-0.0611 [-0.0818, -0.0415]	-0.455 $\rightarrow$ -0.140
ResNet-50	Pos-Weighted BCE	Pneumonia	10.05%	0.0904	0.589 [0.545, 0.629]	0.1137 $\rightarrow$ 0.0982	-0.0155 [-0.0254, -0.0059]	-0.257 $\rightarrow$ -0.086
ResNet-50	Pos-Weighted BCE	Edema	21.33%	0.1678	0.570 [0.538, 0.602]	0.1929 $\rightarrow$ 0.2130	+0.0201 [+0.0092, +0.0313]	-0.149 $\rightarrow$ -0.269

all 16 architecture-loss-pathology combinations with patient-clustered 1,000-replicate bootstrap 95% confidence intervals. As shown, study-level results demonstrate that the principal findings were qualitatively concordant regardless of whether the statistical unit is the individual radiograph or the diagnostic study: unweighted models yield comparable discrimination across views, while external discrimination remains pathology-dependent (AUROC approaching chance for Cardiomegaly), and analytic logit recalibration ($z - \log w$) on positive-weighted models yields statistically significant reductions in study-level Brier error for Cardiomegaly ($\Delta\text{Brier} = -0.0601$ on DenseNet-121 and -0.0642 on ResNet-50) and Pneumonia ($\Delta\text{Brier} = -0.0100$ and -0.0163 , both intervals strictly below zero), alongside consistent point-estimate reductions for Pleural Effusion (-0.0102 and -0.0118).

Section S3: Complete In-Domain Factorial Evaluation Matrix

Table S3 provides the comprehensive 17-column longitudinal evaluation matrix across all models, loss objectives, and thoracic abnormalities across deidentified-date strata T_1 , T_2 , and T_3 , displaying discrimination (AUROC), calibration accuracy (Brier score, ECE), and negative log-likelihood (log-loss) alongside mean and inter-seed standard deviations across random seeds $S \in \{42, 123, 2026\}$.

Table S2: Supplementary Table S2: Diagnostic Study-Level Sensitivity Analysis on the Official MIMIC-CXR-JPG Test Cohort ($N = 3,041$ Studies across $N = 289$ Unique Patients). Multi-image studies are aggregated using maximum probability pooling ($\max_i p_i$). Evaluated on 3-seed probability ensembles with patient-clustered 1,000-replicate bootstrap 95% confidence intervals, local null Brier reference ($\text{Brier}_{\text{null}} = \pi(1 - \pi)$), and paired recalibration contrast ($\Delta\text{Brier} = \text{Brier}_{\text{corr}} - \text{Brier}_{\text{raw}}$).

Architecture	Objective	Pathology	Study Prevalence	Study $\text{Brier}_{\text{null}}$	AUROC [95% CI]	Brier (Raw $\rightarrow$ Corr)	ΔBrier [95% CI]	BSS (Raw $\rightarrow$ Corr)
DenseNet-121	Unweighted BCE	Pleural Effusion	32.55%	0.2196	0.695 [0.663, 0.723]	0.2488	–	-0.133
DenseNet-121	Unweighted BCE	Cardiomegaly	26.54%	0.1951	0.526 [0.497, 0.559]	0.2378	–	-0.219
DenseNet-121	Unweighted BCE	Pneumonia	10.16%	0.0913	0.608 [0.562, 0.650]	0.0996	–	-0.091
DenseNet-121	Unweighted BCE	Edema	21.64%	0.1697	0.565 [0.528, 0.600]	0.2157	–	-0.271
DenseNet-121	Pos-Weighted BCE	Pleural Effusion	32.55%	0.2196	0.674 [0.643, 0.704]	0.2556 $\rightarrow$ 0.2453	-0.0102 [-0.0427, +0.0236]	-0.164 $\rightarrow$ -0.117
DenseNet-121	Pos-Weighted BCE	Cardiomegaly	26.54%	0.1951	0.527 [0.497, 0.558]	0.2933 $\rightarrow$ 0.2332	-0.0601 [-0.0814, -0.0367]	-0.503 $\rightarrow$ -0.195
DenseNet-121	Pos-Weighted BCE	Pneumonia	10.16%	0.0913	0.599 [0.555, 0.639]	0.1091 $\rightarrow$ 0.0992	-0.0100 [-0.0179, -0.0018]	-0.195 $\rightarrow$ -0.086
DenseNet-121	Pos-Weighted BCE	Edema	21.64%	0.1697	0.583 [0.550, 0.616]	0.1937 $\rightarrow$ 0.2165	+0.0228 [+0.0126, +0.0325]	-0.141 $\rightarrow$ -0.275
ResNet-50	Unweighted BCE	Pleural Effusion	32.55%	0.2196	0.682 [0.650, 0.714]	0.2431	–	-0.107
ResNet-50	Unweighted BCE	Cardiomegaly	26.54%	0.1951	0.521 [0.494, 0.548]	0.2340	–	-0.200
ResNet-50	Unweighted BCE	Pneumonia	10.16%	0.0913	0.584 [0.543, 0.624]	0.0992	–	-0.087
ResNet-50	Unweighted BCE	Edema	21.64%	0.1697	0.516 [0.480, 0.553]	0.2158	–	-0.272
ResNet-50	Pos-Weighted BCE	Pleural Effusion	32.55%	0.2196	0.625 [0.594, 0.655]	0.2685 $\rightarrow$ 0.2567	-0.0118 [-0.0427, +0.0251]	-0.223 $\rightarrow$ -0.169
ResNet-50	Pos-Weighted BCE	Cardiomegaly	26.54%	0.1951	0.523 [0.493, 0.552]	0.2858 $\rightarrow$ 0.2217	-0.0642 [-0.0852, -0.0434]	-0.465 $\rightarrow$ -0.136
ResNet-50	Pos-Weighted BCE	Pneumonia	10.16%	0.0913	0.588 [0.542, 0.627]	0.1155 $\rightarrow$ 0.0992	-0.0163 [-0.0267, -0.0068]	-0.265 $\rightarrow$ -0.086
ResNet-50	Pos-Weighted BCE	Edema	21.64%	0.1697	0.571 [0.538, 0.606]	0.1951 $\rightarrow$ 0.2164	+0.0213 [+0.0100, +0.0330]	-0.149 $\rightarrow$ -0.275

Table S3: Supplementary Table S3: Complete 17-Column Longitudinal Factorial Matrix across DenseNet-121 and ResNet-50 Architectures, Loss Objectives, and Target Thoracic Pathologies across Strata T_1 (2015-H2), T_2 (2016), and T_3 (2017). Values represent mean $\pm$ standard deviation across three random seeds.

Architecture	Objective	Target	AUROC T_1	AUROC T_2	AUROC T_3	AUROC Δ	Brier T_1	Brier T_2	Brier T_3	Brier Δ	ECE T_1	ECE T_2	ECE Δ	LogLoss T_1	LogLoss T_2	LogLoss Δ
DenseNet-121	Unweighted BCE	Pleural Effusion	0.9335 $\pm$ 0.0052	0.8895 $\pm$ 0.0115	0.8731 $\pm$ 0.0120	-0.0604	0.0369 $\pm$ 0.0029	0.0356 $\pm$ 0.0009	0.0279 $\pm$ 0.0009	-0.0090	0.0232 $\pm$ 0.0078	0.0090 $\pm$ 0.0022	-0.0143	0.1318 $\pm$ 0.0076	0.1114 $\pm$ 0.0027	-0.0204
DenseNet-121	Unweighted BCE	Cardiomegaly	0.9075 $\pm$ 0.0116	0.8965 $\pm$ 0.0140	0.9017 $\pm$ 0.0170	-0.0057	0.0532 $\pm$ 0.0012	0.0677 $\pm$ 0.0021	0.0677 $\pm$ 0.0028	+0.0145	0.0140 $\pm$ 0.0009	0.0210 $\pm$ 0.0074	+0.0071	0.1798 $\pm$ 0.0072	0.2231 $\pm$ 0.0102	+0.0433
DenseNet-121	Unweighted BCE	Pneumonia	0.8473 $\pm$ 0.0129	0.8818 $\pm$ 0.0166	0.8647 $\pm$ 0.0172	+0.0174	0.0235 $\pm$ 0.0008	0.0182 $\pm$ 0.0003	0.0187 $\pm$ 0.0006	-0.0048	0.0142 $\pm$ 0.0011	0.0095 $\pm$ 0.0016	-0.0046	0.0996 $\pm$ 0.0035	0.0808 $\pm$ 0.0039	-0.0187
DenseNet-121	Unweighted BCE	Edema*	0.5402 $\pm$ 0.0467	0.8640 $\pm$ 0.0686	–	–	0.0021 $\pm$ 0.0000	0.0032 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	-0.0021	0.0029 $\pm$ 0.0013	0.0029 $\pm$ 0.0029	-0.0000	0.0163 $\pm$ 0.0008	0.0029 $\pm$ 0.0029	-0.0134
DenseNet-121	Weighted BCE	Pleural Effusion	0.9061 $\pm$ 0.0537	0.8676 $\pm$ 0.0457	0.8391 $\pm$ 0.0657	-0.0670	0.1981 $\pm$ 0.0394	0.1287 $\pm$ 0.0432	0.1254 $\pm$ 0.0456	-0.0173	0.1064 $\pm$ 0.0702	0.2227 $\pm$ 0.0705	+0.0264	0.3646 $\pm$ 0.1179	0.4100 $\pm$ 0.1279	+0.0454
DenseNet-121	Weighted BCE	Cardiomegaly	0.8729 $\pm$ 0.0540	0.8599 $\pm$ 0.0600	0.8575 $\pm$ 0.0609	-0.0164	0.1262 $\pm$ 0.0247	0.1358 $\pm$ 0.0316	0.1360 $\pm$ 0.0306	+0.0098	0.2031 $\pm$ 0.0507	0.1969 $\pm$ 0.0486	-0.0062	0.3962 $\pm$ 0.0807	0.4241 $\pm$ 0.0906	+0.0259
DenseNet-121	Weighted BCE	Pneumonia	0.8181 $\pm$ 0.0206	0.8494 $\pm$ 0.0313	0.8424 $\pm$ 0.0406	+0.0243	0.1313 $\pm$ 0.0533	0.1308 $\pm$ 0.0555	0.1298 $\pm$ 0.0513	-0.0015	0.2393 $\pm$ 0.0894	0.2425 $\pm$ 0.0883	+0.0032	0.4226 $\pm$ 0.1511	0.4166 $\pm$ 0.1471	-0.0000
DenseNet-121	Weighted BCE	Edema*	0.6127 $\pm$ 0.0268	0.8753 $\pm$ 0.0129	–	–	0.0614 $\pm$ 0.0637	0.0672 $\pm$ 0.0763	0.0638 $\pm$ 0.0754	-0.0022	0.1191 $\pm$ 0.1228	0.1274 $\pm$ 0.1327	+0.0082	0.2605 $\pm$ 0.2621	0.2603 $\pm$ 0.2388	-0.0002
ResNet-50	Unweighted BCE	Pleural Effusion	0.9295 $\pm$ 0.0113	0.8755 $\pm$ 0.0293	0.8608 $\pm$ 0.0190	-0.0687	0.0372 $\pm$ 0.0022	0.0364 $\pm$ 0.0013	0.0284 $\pm$ 0.0010	-0.0088	0.0162 $\pm$ 0.0050	0.0132 $\pm$ 0.0030	-0.0029	0.1294 $\pm$ 0.0099	0.1145 $\pm$ 0.0067	-0.0149
ResNet-50	Unweighted BCE	Cardiomegaly	0.8898 $\pm$ 0.0177	0.8756 $\pm$ 0.0317	0.8785 $\pm$ 0.0436	-0.0112	0.0572 $\pm$ 0.0037	0.0709 $\pm$ 0.0059	0.0712 $\pm$ 0.0051	+0.0140	0.0233 $\pm$ 0.0190	0.0175 $\pm$ 0.0081	-0.0058	0.1954 $\pm$ 0.0193	0.2379 $\pm$ 0.0174	+0.0425
ResNet-50	Unweighted BCE	Pneumonia	0.8432 $\pm$ 0.0178	0.8659 $\pm$ 0.0130	0.8798 $\pm$ 0.0169	+0.0366	0.0242 $\pm$ 0.0007	0.0187 $\pm$ 0.0003	0.0191 $\pm$ 0.0003	0.0051	0.0128 $\pm$ 0.0050	0.0106 $\pm$ 0.0035	-0.0023	0.1021 $\pm$ 0.0059	0.0809 $\pm$ 0.0035	-0.0215
ResNet-50	Unweighted BCE	Edema*	0.5564 $\pm$ 0.0368	0.8966 $\pm$ 0.0216	–	–	0.0021 $\pm$ 0.0000	0.0032 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	-0.0021	0.0023 $\pm$ 0.0005	0.0017 $\pm$ 0.0009	-0.0006	0.0175 $\pm$ 0.0010	0.0017 $\pm$ 0.0009	-0.0158
ResNet-50	Weighted BCE	Pleural Effusion	0.8769 $\pm$ 0.0508	0.8467 $\pm$ 0.0458	0.8191 $\pm$ 0.0496	-0.0598	0.1236 $\pm$ 0.0325	0.1417 $\pm$ 0.0277	0.1392 $\pm$ 0.0310	-0.0187	0.2481 $\pm$ 0.0733	0.2754 $\pm$ 0.0769	+0.0270	0.4003 $\pm$ 0.0885	0.4365 $\pm$ 0.0750	+0.0361
ResNet-50	Weighted BCE	Cardiomegaly	0.8298 $\pm$ 0.0550	0.8646 $\pm$ 0.0701	0.7990 $\pm$ 0.0788	-0.0508	0.1415 $\pm$ 0.0124	0.1526 $\pm$ 0.0083	0.1522 $\pm$ 0.0085	-0.0007	0.2450 $\pm$ 0.0348	0.2338 $\pm$ 0.0258	-0.0112	0.4427 $\pm$ 0.0389	0.4678 $\pm$ 0.0505	+0.0251
ResNet-50	Weighted BCE	Pneumonia	0.8349 $\pm$ 0.0199	0.8359 $\pm$ 0.0345	0.8233 $\pm$ 0.0533	-0.0116	0.1667 $\pm$ 0.0584	0.1681 $\pm$ 0.0583	0.1691 $\pm$ 0.0613	+0.0024	0.3189 $\pm$ 0.1175	0.3253 $\pm$ 0.1208	+0.0064	0.5656 $\pm$ 0.1443	0.5096 $\pm$ 0.1527	+0.0560
ResNet-50	Weighted BCE	Edema*	0.5490 $\pm$ 0.0852	0.8310 $\pm$ 0.0437	–	–	0.0693 $\pm$ 0.0461	0.0749 $\pm$ 0.0509	0.0746 $\pm$ 0.0494	+0.0052	0.1545 $\pm$ 0.1121	0.1670 $\pm$ 0.1169	+0.0126	0.2193 $\pm$ 0.1374	0.2295 $\pm$ 0.1409	+0.0102

Section S4: Complete Multi-Calibrator Parameter Matrix

Table S4 documents the exhaustive 12-column multi-calibrator evaluation on Pleural Effusion across 3-seed probability ensembles on held-out deidentified-date strata T_1 , T_2 , and T_3 , including Cox calibration regression intercepts (α), calibration slopes (β), ECE, log-loss, and patient-clustered 2,000-replicate bootstrap contrasts against uncalibrated models.

Table S4: Supplementary Table S4: Complete 12-Column Multi-Calibrator Comparison on Pleural Effusion across 3-Seed Probability Ensembles, Evaluating Cox Intercept (α), Slope (β), ECE, Log-Loss, and Paired Bootstrap Contrasts.

Architecture	Loss Objective	Calibrator	T1 Brier	T2 Brier	T3 Brier	Δ Brier (T3 - T1)	T3 Intercept (α)	T3 Slope (β)	T3 ECE	T3 Log-Loss	T3 Δ Brier vs Raw	95% Bootstrap CI
DenseNet-121	Weighted BCE	Raw (Uncalibrated)	0.0920	0.1111	0.1079	+0.0158	-2.9638	1.0484	0.2227	0.3572	-	-
DenseNet-121	Weighted BCE	Temperature Scaling (T*)	0.0911	0.1100	0.1067	+0.0157	-2.9615	1.0289	0.2190	0.3540	-0.0011	[-0.0014, -0.0009]
DenseNet-121	Weighted BCE	Analytic Offset ($z - \log w$)	0.0359	0.0348	0.0284	-0.0075	-0.4451	0.9292	0.0126	0.1134	-0.0795	[-0.0803, -0.0697]
DenseNet-121	Weighted BCE	Fitted Offset ($z + a^*$)	0.0354	0.0352	0.0289	-0.0066	-0.6013	0.9423	0.0183	0.1148	-0.0790	[-0.0885, -0.0696]
DenseNet-121	Weighted BCE	Platt Scaling ($a + bz$)	0.0357	0.0345	0.0281	-0.0076	-0.2288	1.1280	0.0176	0.1136	-0.0798	[-0.0894, -0.0702]
DenseNet-121	Unweighted BCE	Raw (Uncalibrated)	0.0356	0.0341	0.0268	-0.0088	+0.1471	1.0439	0.0040	0.1054	-	-
DenseNet-121	Unweighted BCE	Temperature Scaling (T*)	0.0350	0.0339	0.0268	-0.0082	+0.1528	1.1735	0.0082	0.1066	+0.0001	[-0.0003, +0.0004]
DenseNet-121	Unweighted BCE	Fitted Offset ($z + a^*$)	0.0333	0.0343	0.0274	-0.0059	-0.4177	1.0422	0.0160	0.1089	+0.0007	[-0.0006, +0.0018]
DenseNet-121	Unweighted BCE	Platt Scaling ($a + bz$)	0.0333	0.0355	0.0285	-0.0048	-0.7251	0.8828	0.0175	0.1102	+0.0017	[-0.0000, +0.0034]
ResNet-50	Weighted BCE	Raw (Uncalibrated)	0.1101	0.1254	0.1227	+0.0126	-2.8526	1.9498	0.2751	0.4015	-	-
ResNet-50	Weighted BCE	Temperature Scaling (T*)	0.0876	0.1045	0.1008	+0.0131	-2.6080	1.1840	0.1993	0.3187	-0.0220	[-0.0241, -0.0198]
ResNet-50	Weighted BCE	Analytic Offset ($z - \log w$)	0.0402	0.0367	0.0291	-0.0111	+1.0751	1.4052	0.0158	0.1149	-0.0936	[-0.1022, -0.0844]
ResNet-50	Weighted BCE	Fitted Offset ($z + a^*$)	0.0384	0.0361	0.0291	-0.0093	+0.7008	1.5504	0.0229	0.1208	-0.0936	[-0.1016, -0.0851]
ResNet-50	Weighted BCE	Platt Scaling ($a + bz$)	0.0380	0.0364	0.0297	-0.0082	-0.1911	1.1777	0.0223	0.1211	-0.0930	[-0.1007, -0.0849]
ResNet-50	Unweighted BCE	Raw (Uncalibrated)	0.0361	0.0351	0.0272	-0.0088	-0.0766	1.0945	0.0109	0.1087	-	-
ResNet-50	Unweighted BCE	Temperature Scaling (T*)	0.0361	0.0351	0.0273	-0.0088	-0.0957	1.0974	0.0107	0.1089	+0.0000	[-0.0001, +0.0001]
ResNet-50	Unweighted BCE	Fitted Offset ($z + a^*$)	0.0352	0.0354	0.0277	-0.0074	-0.3631	1.0755	0.0189	0.1113	+0.0005	[-0.0001, +0.0011]
ResNet-50	Unweighted BCE	Platt Scaling ($a + bz$)	0.0351	0.0363	0.0284	-0.0066	-0.7099	0.8891	0.0204	0.1124	+0.0012	[-0.0001, +0.0024]

Section S5: Complete Selective Prediction Workload Triage Matrix

Table S5 provides the complete 24-row selective prediction and human-in-the-loop referral simulation on later held-out stratum T_3 ($N = 2,436$ radiographs, 85 true positive Pleural Effusion cases) across coverage deciles from 100% down to 50%, detailing automated decisions, referred cases, retained true and false positives, retained sensitivity, full-cohort system triage sensitivity, retained AUROC, and retained Brier score.

Section S6: Reproducibility and Code Execution Guide

The complete codebase is organized to support deterministic replication from raw prediction archives through all tables, figures, and statistical tests:

1. **Environment Setup:** Python 3.10+ with PyTorch, torchvision, scikit-learn, scipy, pandas, numpy, and matplotlib.
2. **Stage 3 Recalibration Execution:**

python3 scripts/run_stage3_recalibration.py

Table S5: Supplementary Table S5: Complete 24-Row Selective Prediction and Reader Referral Triage Simulation on Later Held-Out Stratum T_3 ($N = 2,436$ Radiographs, 85 Positive Pleural Effusion Cases) across All Coverage Regimes (100% to 50%).

Coverage	Strategy	Automated Decisions (N)	Referred to Reader (N)	Retained Positives (N)	Automated True Positives (N)	Automated False Negatives (Missed Cases)	Retained Sensitivity	Full Cohort Triage Sensitivity	Retained AUROC	Retained Bias
100%	Weighted BCE (Raw)	2,436	0 (0%)	85	69	16.0	0.8118	0.8118	0.8691	0.1079
100%	Weighted BCE (Analytic Corrected)	2,436	0 (0%)	85	69	16.0	0.8118	0.8118	0.8623	0.0284
100%	Unweighted BCE (Raw)	2,436	0 (0%)	85	73	12.0	0.8588	0.8588	0.8896	0.0288
100%	Random Deferral Baseline	2,436	0 (0%)	85.0	69.0	16.0	0.8118	0.8118	0.8623	0.1079
90%	Weighted BCE (Raw)	2,193	243 (9%)	82	68	14.0	0.8203	0.8203	0.8789	0.1062
90%	Weighted BCE (Analytic Corrected)	2,193	243 (9%)	79	64	15.0	0.8101	0.8235	0.8735	0.0289
90%	Unweighted BCE (Raw)	2,193	243 (9%)	78	66	12.0	0.8462	0.8588	0.8923	0.0266
90%	Random Deferral Baseline	2,193	243 (9%)	78.5	62.0	14.5	0.8106	0.8295	0.8895	0.1080
80%	Weighted BCE (Raw)	1,949	487 (19%)	77	67	10.0	0.8701	0.8824	0.8882	0.1012
80%	Weighted BCE (Analytic Corrected)	1,949	487 (19%)	67	52	15.0	0.7761	0.8225	0.8896	0.0264
80%	Unweighted BCE (Raw)	1,949	487 (19%)	69	57	12.0	0.8261	0.8588	0.9027	0.0256
80%	Random Deferral Baseline	1,949	487 (19%)	68.0	55.0	13.0	0.8090	0.8472	0.8890	0.1080
70%	Weighted BCE (Raw)	1,706	730 (39%)	66	59	7.0	0.8939	0.9176	0.8997	0.0956
70%	Weighted BCE (Analytic Corrected)	1,706	730 (39%)	44	31	13.0	0.7045	0.8471	0.8876	0.0178
70%	Unweighted BCE (Raw)	1,706	730 (39%)	50	39	11.0	0.7869	0.8796	0.9051	0.0194
70%	Random Deferral Baseline	1,706	730 (39%)	50.5	48.2	11.3	0.8699	0.8669	0.9081	0.1081
60%	Weighted BCE (Raw)	1,462	974 (49%)	56	51	5.0	0.9107	0.9412	0.9058	0.0866
60%	Weighted BCE (Analytic Corrected)	1,462	974 (49%)	20	11	9.0	0.5560	0.8941	0.8176	0.0095
60%	Unweighted BCE (Raw)	1,462	974 (49%)	29	22	7.0	0.7586	0.9176	0.8788	0.0120
60%	Random Deferral Baseline	1,462	974 (49%)	51.0	41.4	9.6	0.8122	0.8873	0.9083	0.1080
50%	Weighted BCE (Raw)	1,218	1,218 (50%)	30	45	5.0	0.9000	0.9412	0.9093	0.0834
50%	Weighted BCE (Analytic Corrected)	1,218	1,218 (50%)	7	2	5.0	0.2857	0.9412	0.5572	0.0051
50%	Unweighted BCE (Raw)	1,218	1,218 (50%)	14	7	7.0	0.3900	0.9176	0.8954	0.0080
50%	Random Deferral Baseline	1,218	1,218 (50%)	42.5	34.4	8.1	0.8101	0.9051	0.8954	0.1078

Computes probability ensembles ($\bar{p}_m = \frac{1}{3} \sum_{s=1}^3 p_{m,s}$) and ensemble logit offsets ($\Delta z = \frac{1}{3} \sum_{s=1}^3 (z_{w,s} - z_{u,s})$) across all 8,302 held-out radiographs (16,604 paired image-architecture evaluations, alongside all 49,812 seed-specific pairs) for raw, temperature-scaled, analytic offset ($z - \log w$), fitted offset ($z + a^*$), and Platt-scaled models, running 2,000-replicate patient-clustered bootstrap inference.

3. Publication Artifact Compilation:

`python3 scripts/compile_publication_artifacts.py`

Generates Tables 1–4 in LaTeX and markdown formats, alongside publication figures (Figures 2–6).

4. Verification Audit:

`python3 scripts/audit_cibm.py`

Verifies all formatting, word count, highlight length (≤ 85 characters), and statutory compliance rules for Computers in Biology and Medicine.